

Top 10 Key Insights: Credit Card Fraud Analysis

A deep dive into 555,719 transactions — uncovering fraud patterns, high-risk segments, and prevention opportunities across categories, geographies, and merchant networks.



Insight 1: Extremely Low Fraud Rate, Extreme Imbalance

0.39%

Fraud Rate

2,145 fraud cases out of 555,719 total transactions

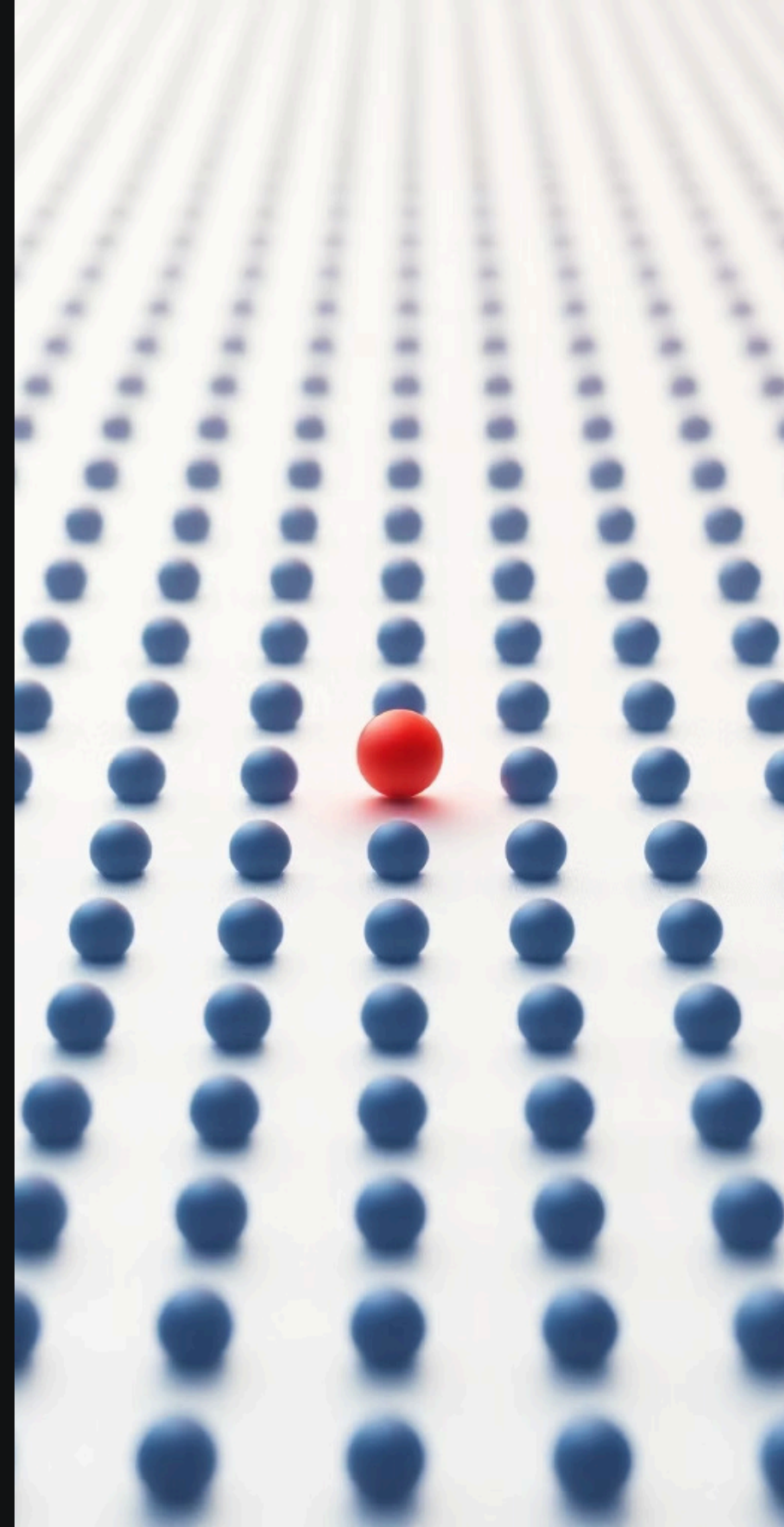
259:1

Imbalance Ratio

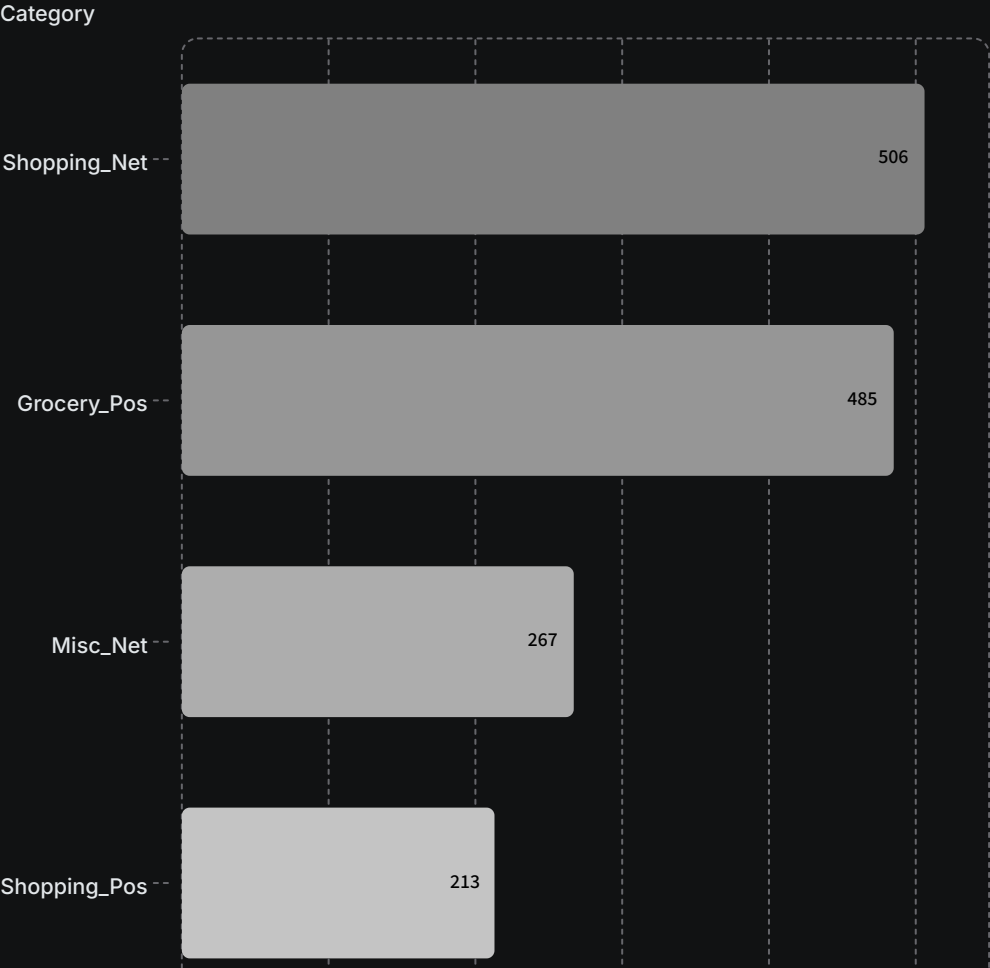
Normal vs. fraudulent transactions

Why This Matters

Severe class imbalance means standard ML algorithms default to predicting "normal" for every transaction — and still achieve 99.6% accuracy. Proper rebalancing techniques like **SMOTE** or cost-sensitive learning are essential before any model training.



Insight 2: Shopping Categories Are Primary Fraud Targets



Key Takeaway

Online shopping platforms account for the highest fraud volume. **Shopping_Net alone carries a 1.21% fraud rate** — the highest of any category. Online channels are approximately **3x more vulnerable** than physical point-of-sale environments.

📄 Prioritise real-time scoring rules for all net/online merchant category codes.

Insight 3 & 4: Demographics vs. Geography

Gender: A Non-Predictor

Near-identical rates confirm gender carries no predictive signal. Redirect model feature budget toward behavioural and transactional attributes.

Female

1,164 cases · **0.38% rate** 304,886 transactions

→ **New York — 175 cases**

→ **Texas — 113 cases**

Top Fraud States

Top 5 states represent ~25% of all fraud — population-dense corridors warrant heightened monitoring.

Male

981 cases · **0.39% rate** 250,833 transactions

→ **Pennsylvania — 114 cases**

→ **Illinois & California — 76 cases each**



Insight 5: City-Level Fraud Hotspots Identified

Camden

27 fraud cases — highest city concentration nationwide

Birmingham

25 fraud cases — consistent pattern across multiple merchants

Burrton

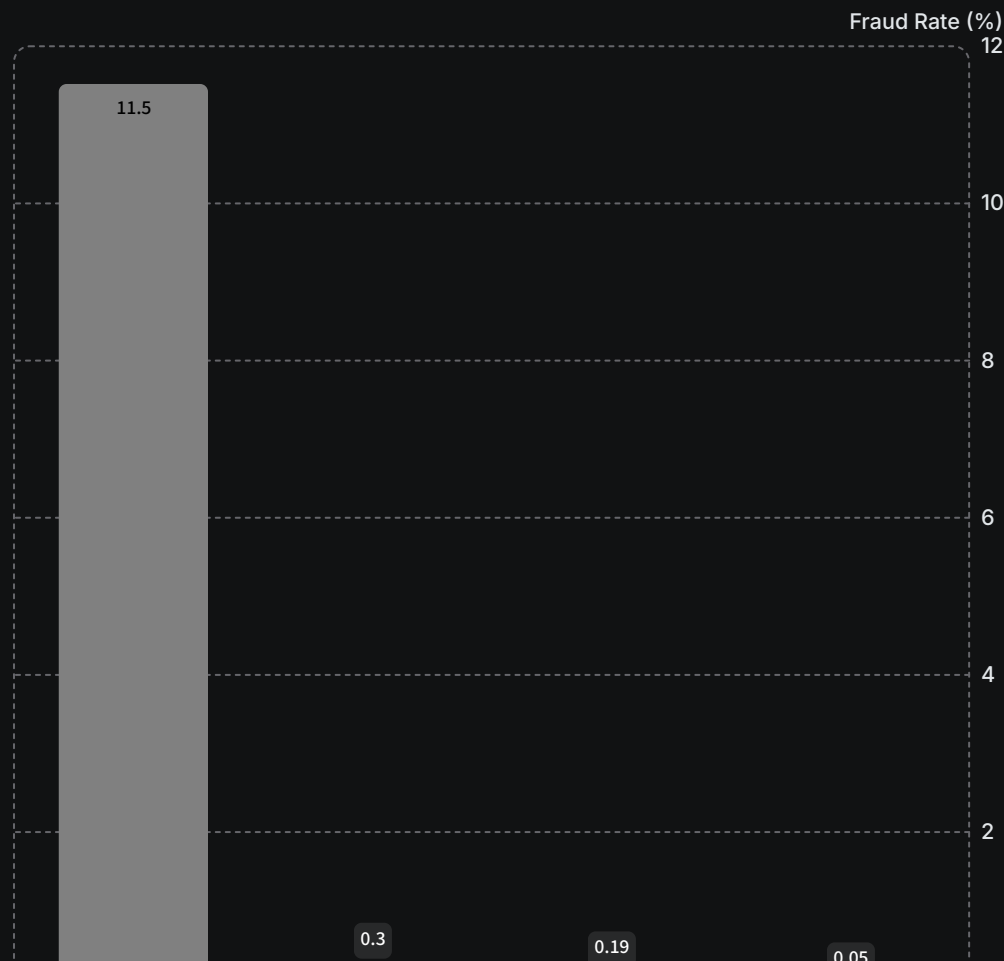
19 fraud cases — disproportionate given population size

Clarks Mills

18 fraud cases — small city with outsized fraud density

These cities show fraud rates far exceeding their transaction share — a signal of localised compromise or organised activity.

Insight 6: Transaction Amount Is the Strongest Fraud Indicator



230x Risk Differential

Very large transactions carry an **11.52% fraud rate** — 230 times higher than medium-sized transactions at 0.05%. Transaction amount should be a **top-weighted feature** in any fraud scoring model.

📌 Flag all very-large transactions for secondary review or step-up authentication.

Insights 7 & 8: Occupation and Merchant Risk Patterns

High-Risk Occupations

These professions show **10–15x the average fraud rate**, possibly reflecting lifestyle spend patterns or targeted social engineering.

Horticultural Consultant

5.63% fraud rate

Chartered Accountant

5.53% fraud rate

TV Camera Operator

5.36% fraud rate

Designer (TV/Film)

4.85% fraud rate

Romaguera-Cruickshank & Green

2.17% fraud rate —
highest flagged
merchant

Lemke-Gutmann

2.15% fraud rate

Mosciski-Ziemann & Farrell

2.07% fraud rate

Heathcote-Yost & Kertzmann

2.05% fraud rate

Compromised Merchants

Fraud rates **5–6x the baseline** indicate potential POS compromise or insider risk at these merchants.

Insight 9: ZIP Code Clustering Reveals Fraud Rings



ZIP 12037

6.61% (16 cases / 242 transactions)



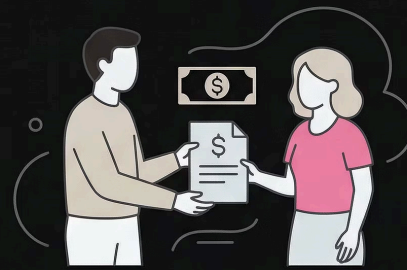
ZIP 32565

6.61% (16 cases / 242 transactions)



ZIP 49629

6.51% (14 cases / 215 transactions)



ZIP 68438

6.40% (13 cases / 203 transactions)

Organised Fraud Signature

These four ZIP codes each show fraud rates exceeding **6.4%** — roughly 16x the dataset average of 0.39%. The tight geographic clustering within small transaction pools strongly suggests **coordinated fraud ring activity**, not random isolated events.

Recommended Action

Apply network-based graph analysis to surface shared attributes — device IDs, IP ranges, or merchant overlap — across these ZIP clusters.



Insight 10: Massive Financial Impact — and a Clear Prevention Opportunity

\$38.56M

Total Transaction Volume

\$1.13M

Total Fraud Loss

2.94% of total transaction value

\$527

Avg. Fraud Amount

Per fraudulent transaction

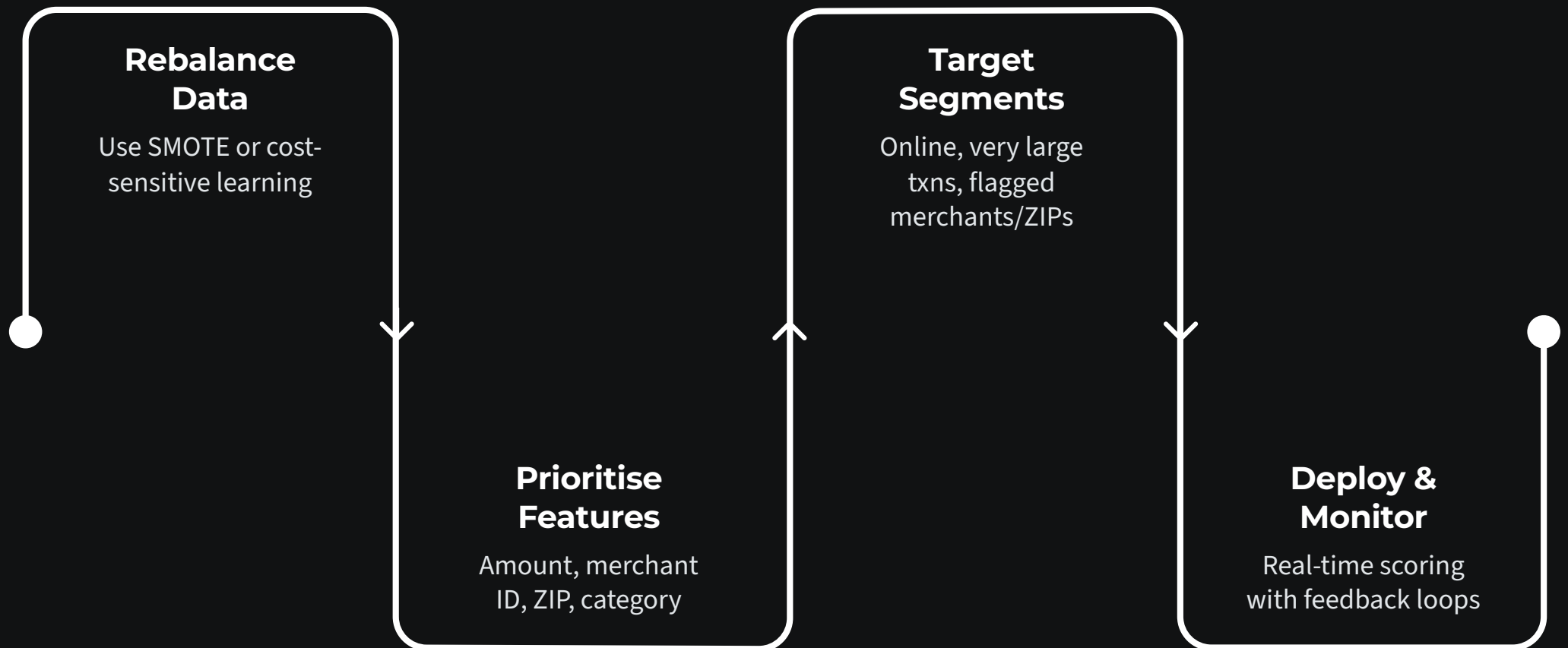
81.33%

ROI Potential

Return on fraud prevention investment

At an 80% detection rate, an estimated **\$906,000 in annual savings** is achievable — making a robust ML-driven fraud detection system a high-return investment with a compelling business case.

Summary: Actionable Priorities for Fraud Teams



These ten insights translate directly into a prioritised model-building and operations roadmap — from data preparation through to real-time deployment — with an estimated **\$906K annual fraud prevention opportunity** at stake.